RCE-2 v6.14

Observational Causal Effect Estimation — No Ground Truth Required

v6.14 replaces the need for interventional queries (which require the ground truth compute_fn) with observational estimation methods: Propensity Score Matching, Inverse Probability Weighting, and Doubly-Robust Estimation. The substrate can now estimate causal effects from past computation logs alone — no offline interventional training required. This makes causal discovery practical in production.

Project	Omega-X v6.14
Document	RCE-2 v6.14 Observational Causal Estimation Report
Date	2026-07-16
Status	[MEASURED] [IMPLEMENTED] [TESTED]
Test Suite	197 tests (182 prior + 15 observational), all passing
Methods	PSM + IPW + Doubly-Robust (cross-checked)
Key Result	Causal effects estimated without ground truth compute_fn

1. Executive Summary

This report documents the v6.14 Observational Causal Effect Estimation layer. The v6.11-v6.13 causal layers require INTERVENTIONAL queries (do-operator) to discover causal features. This requires the ground truth compute_fn — the actual computation the substrate is trying to avoid. In production, we don't have compute_fn available for offline interventional queries; we only have OBSERVATIONAL data (features, outcomes) from past computations.

The v6.14 ObservationalCausalEstimator implements three methods from the causal inference literature: Propensity Score Matching (PSM), Inverse Probability Weighting (IPW), and Doubly-Robust Estimation (DR). All three estimate causal effects from observational data alone — no ground truth compute_fn required. The substrate cross-checks all three methods: a feature is classified as causal only if all methods agree on the direction of the effect.

Key Results

Metric	Value	Notes
PSM identifies causal features	Verified	ACE > 0.5 for true causes
IPW identifies causal features	Verified	Cross-checks PSM
DR identifies causal features	Verified	Most reliable (doubly robust)
Irrelevant features excluded	Verified	ACE < threshold for non-causes
Methods cross-checked	Verified	Causal only if all agree
No compute_fn required	Verified	Pure observational estimation
Test suite	197 tests	15 new observational tests, all passing

2. Three Estimation Methods

The ObservationalCausalEstimator implements three complementary methods from the causal inference literature. Each estimates the Average Causal Effect (ACE) from observational data using different assumptions and techniques.

2.1 Propensity Score Matching (PSM)

```
Propensity score: e(x) = P(T=1 | X=x)
  Estimated via logistic regression on confounders

PSM estimator:
  For each treated unit (T=1), find control unit (T=0)
  with most similar propensity score
  ACE_PSM = mean[Y_treated] - mean[Y_matched_control]
```

2.2 Inverse Probability Weighting (IPW)

```
IPW estimator:
  Weight each observation by 1/propensity_score
  Creates pseudo-population where T is independent of X

  ACE_IPW = mean[T*Y/e(X)] - mean[(1-T)*Y/(1-e(X))]
```

2.3 Doubly-Robust Estimation (DR)

```
DR estimator (combines propensity + outcome regression):
  m1(x) = E[Y | T=1, X=x]  (outcome regression for treated)
```

```
m0(x) = E[Y | T=0, X=x]  (outcome regression for control)

ACE_DR = mean[m1(X) - m0(X)]
       + mean[T*(Y-m1(X))/e(X)]
       - mean[(1-T)*(Y-m0(X))/(1-e(X))]

Doubly robust: consistent if EITHER propensity OR outcome
model is correct. Most reliable of the three methods.
```

3. Honest Limitations

Observational causal estimation has a fundamental limitation: it cannot distinguish confounders from causes when they are perfectly correlated. If feature A causes both feature B and the outcome, then B correlates perfectly with the outcome but does not cause it. Observational methods will estimate ACE(B) > 0 because intervening on B (in the observational data) is associated with outcome changes — but this is confounding, not causation.

3.1 The Confounder Problem

```
True structure: A -> B, A -> Y (B is confounder)
Observational data: B and Y are perfectly correlated

Observational estimate:
  ACE(A) = 1.0 (correct — A is causal)
  ACE(B) = 1.0 (WRONG — B is confounder, not cause)

Interventional estimate (v6.11):
  ACE(A) = 1.0 (correct)
  ACE(B) = 0.0 (correct — intervening on B doesn't change Y)
```

The substrate documents this limitation honestly: observational estimation is a FIRST PASS that identifies candidate causal features. Interventional queries (v6.11) are needed for validation when compute_fn is available. The hybrid approach (v6.12) combines both: observational PC for structure, interventional ACE for validation.

3.2 Assumptions

All three methods assume: (1) no unobserved confounders (ignorability), (2) overlap/positivity ($0 < P(T=1|X) < 1$ for all X), (3) SUTVA (one unit's treatment doesn't affect another's outcome). Violations of these assumptions produce biased estimates. The substrate cannot verify these assumptions from data alone — they are domain knowledge that the caller must provide.

4. Conclusion

v6.14 adds observational causal estimation to the substrate. The ObservationalCausalEstimator implements PSM, IPW, and DR methods, enabling causal effect estimation from past computation logs alone — no ground truth compute_fn required. This makes causal discovery practical in production: collect (features, outcome) pairs, run the estimator, identify candidate causal features.

The substrate now has both observational and interventional causal discovery:

```
Observational (v6.14): PSM + IPW + DR
  - No compute_fn required
  - First-pass identification
  - Cannot distinguish confounders from causes

Interventional (v6.11): do-operator
```

```
  - Requires compute_fn
  - Gold standard
  - Correctly excludes confounders

Hybrid (v6.12): PC algorithm + ACE supplementation
  - Combines both approaches
  - Most reliable
```

The complete causal reasoning hierarchy now has 8 levels:

```
Level 0: States              (v6.5)
Level 1: Correlational Laws  (v6.5)
Level 2: Refined Laws        (v6.9)
Level 3: Transferred Laws    (v6.10)
Level 4: Causal Laws         (v6.11)
Level 5: Structural Causal   (v6.12)
Level 6: Counterfactual      (v6.13)
Level 7: Observational Causal (v6.14)
Level 8: Online Causal       (future)
```

What v6.14 Does NOT Claim

v6.14 does not claim observational estimates are as reliable as interventional ones. They are not — confounders cannot be detected observationally. v6.14 provides a FIRST PASS that is better than no causal analysis at all, but should be validated with interventional queries when possible. v6.14 also does not claim the propensity score model is correct. Logistic regression is a simple model; more sophisticated models (random forests, neural networks) could improve estimation. These are candidates for v6.15.

Appendix A. File Inventory

Artifact	Location	Notes
Observational estimator	rce/rce2_observational_causal.py	PSM + IPW + DR
v6.14 tests	tests/test_rce2_observational_causal.py	15 tests, all passing
This report	rce/RCE2_V614_REPORT.pdf	PDF

Appendix B. How to Reproduce

```
# Run all 197 tests
cd /home/z/my-project/download
python -m pytest tests/test_rce2.py tests/test_rce2_adversarial.py \
  tests/test_rce2_substrate.py tests/test_rce2_density_gate.py \
  tests/test_rce2_adaptive_gate.py tests/test_rce2_reuse_classifier.py \
  tests/test_rce2_law_refiner.py tests/test_rce2_meta_learner.py \
  tests/test_rce2_causal_discovery.py tests/test_rce2_scm_discovery.py \
  tests/test_rce2_counterfactual.py tests/test_rce2_observational_causal.py

# Quick self-test
python -c "from rce.rce2_observational_causal import self_test; self_test()"
```